



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering.
End Semester Examination
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: CAD/CAM & Robotics.

Course Code: PCC-PE404

Date: 22.11.2025

Time: 13:30 to 15:30

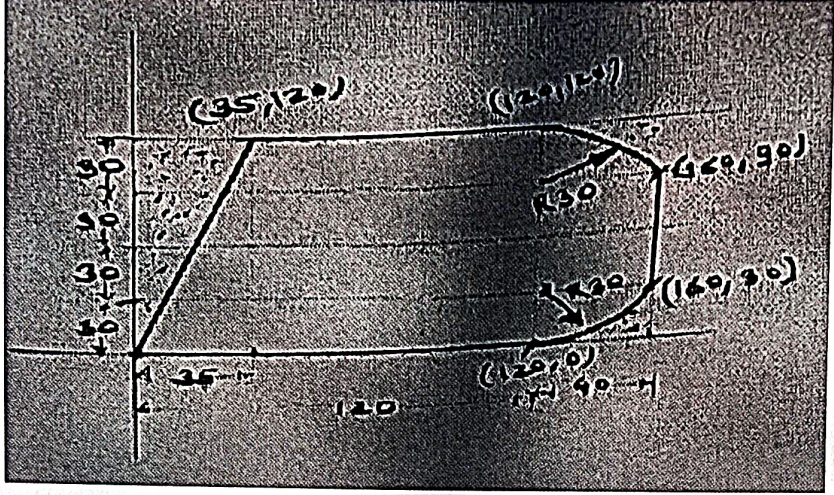
Max. Marks: 60

Note:

Reg. No:

- Assume suitable data whenever necessary and state it clearly.
- Solve **any FIVE questions** from questions 1 to 6.
- Figures to right indicate full marks.
- Paper containing **2 pages**.

Q.N.			Marks	CO	BT
Q.1	a	Translate a square ABCD, A (0,0), B (3,0), C (3,3) and D (0,3) by 2 times in both direction and the scale it by 1.5 times in x-direction and 0.5 times in y-direction. Determine the resultant co-ordinates of polygon.	06	CO2	BT3
	b	What are the limitations found in the general wireframe modelling systems? Explain with an example.	06	CO1	BT1
Q.2	a	Explain the concept of the three basic Boolean operations used in solid modelling. Give neat sketches showing the effect of these operators on any two basic primitives.	06	CO2	BT3
	b	What are the applications of industrial robot? Discuss in brief the types of robots.	06	CO5	BT2
Q.3	a	Difference between NC and CNC machines. (any 12 points)	06	CO3	BT2
	b	Define the "Group technology" and its importance in plant layout and manufacturing process planning.	06	CO4	BT4
Q.4	a	What is automation? Discuss types of automation along with limitations.	06	CO4	BT2
	b	Explain direct numerical control (DNC) with its merits and demerits.	06	CO3	BT2
Q.5	a	Explain different components of a robot.	06	CO5	BT2
	b	Explain flexible manufacturing system with its merits and demerits.	06	CO4	BT4
Q.6	a	Difference between scalar and vector. (any 4 points)	02	CO2	BT2

	As shown in fig, a 10mm thick aluminium block has to be profile milled from outside. Part is fixed in such a way that its top surface is 10mm above machining table. For fixing the part, three holes are already drilled. Before machining initial co-ordinates are $x=0$, $y=-50$ and $z=10$. A 10mm diameter (5mm radius) each mill (milling cutter) with 4 teeth is used. Feed for this operation is 40mm/min and spindle speed is 1000rpm. With help of absolute programming write NC part program for the operation. Shaded portion in machining area to remove. b 	10	CO5	BT5
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== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Learn CAD hardware and software for variety of applications
CO2	Compare different alternative facilities in CAD software
CO3	Implement the NC part programming and its application
CO4	Understand the different concepts in manufacturing system and its application for industrial automation.
CO5	Describe the robot and its different industrial applications.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Academic Year: 2025-26 (semester I)
End Semester Examination

Class: Final Year B.Tech.

Course Name: Tools for Six Sigma
Quality

Course Code: PCC-PE403

Date: 20.11.2025

Time: 13:30 to 15:30

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.1 Answer the following subquestions.

08M

- (a) Define the terms (i) Hidden quality cost (ii) OC curve.
(b) What are the objectives of quality cost?

4M.....CO1BT1

4M.....CO4BT2

Q.2 Solve any two numericals / analysis from the following.

12M

- (a) Mr. X has got the business of printing visiting cards. Each order is considered as a unit. Each order has four defect opportunities i.e. incorrect, typo, damaged or incomplete. Fifty orders have been randomly selected and inspected. Out of the fifty orders, six orders have a problem, and there are a total of seven defects out of fifty orders that have been sampled. Find the DPMO and the sigma level by F. W. Brefogle's equation.

6M.....CO3BT5

- (b) For a PQR Sandwich hotel the following Voice of Customer priority is given. Prioritize the KPIV based on the KPOV.

6M.....CO4BT2

Rating of importance to customer >>		9	6	4	8	
	Process step and inputs	Taste	Construction	On time	Quantity	Total
1	Set up Sandwich maker	9	9	9	3	
2	Set up area	1	1	3	1	
3	Standard (work instruction)	9	9	9	9	
4	Consumers requirements	9	3	3	9	
5	Set up Lighted area	3	9	3	3	
6	Sandwich maker	9	9	9	3	
7	Cup measure	9	9	0	9	
8	Apply the Peanut butter	9	9	0	3	
9	End users requirements	9	9	1	3	
10	Construction tools	9	9	3	9	
11	Lighted area	3	3	3	9	

- (c) What are the key steps for conducting ANOVA analysis in Minitab?

6M.....CO4BT4

Q.3 Attempt any five questions from the following questions.

40M

- (a) Draw the Kano model. Explain how the Kano model helps the businesses to improve its market share and goodwill?

8M.....CO6BT3

- (b) How many phases are there in Quality Function Deployment? How these phases are interlinked with each other.

8M.....CO6BT2

(c) Discuss Taguchi's philosophy for quality improvement.
Suppose a company manufactures widgets, and the target weight for these widgets is 500 grams ($m = 500$). The company has a tolerance range of ± 10 grams, meaning any widget with a weight between 490 and 510 grams is considered acceptable. Widgets outside this specification limit (acceptable limits) are rejected, costing the company \$50 per rejected widget. Calculate the financial loss for a widget with a weight of 509 grams.

8M.....CO5BT5

(d) Discuss the signal-to-noise ratio. How is it used in the Taguchi method?

8M.....CO5BT3

(e) What is orthogonal array? Describe L_8 OA.

8M.....CO5BT4

(f) What are the readings on SFMEA? Explain with a suitable example.

8M.....CO3BT6

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 – Remember / Define, BT2 – Understand / Interpret, BT3 – Apply / Utilize, BT4 – Analyze / Inference,
BT5 – Evaluate / Conclude, BT6 - Create / Elaborate / Formulate

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand History, Concepts, Definitions, Levels of Six Sigma, Characteristics, and objectives of Six Sigma.
CO 2	Learn Road map for six sigma, DMAIC methodology.
CO 3	Use graphical tools, process mapping, FMEA, Ishikawa diagram.
CO 4	Solve numerical based on CI, ANOVA, HT, Correlation and Regression, cost of poor quality, identify prioritize and select six sigma projects.
CO 5	Solve numerical for parameter optimization using concepts of DOE, Robust design, Response plots, O.A., S/N ratios.
CO 6	Design a product using concept of QFD, and House of Quality.



SGGS INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED – 431 606
Department of Production Engineering
Academic Year 2024-2025

Dt: 25 /11/25
 13.30-15.30

ISE-2 Examination Class: B.Tech. (Prod)
Subject: Productivity improvement techniques (PCC-PE-402)
***CO: Course Outcomes.**

Max Marks: 40

NOTE: SOLVE ANY SIX QUESTIONS FROM THE FOLLOWING.

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Q.1	Draw a multiple activity chart recording worker and machine for finish mill casting product on a vertical milling machine and compute the cycle time, worker utilization (%) and machine utilization (%). The process details are :Worker removes finished casting; cleans with compressed air(0.2min),gauges depth on surface plate (0.2min),breaks sharp edge with file; cleans with compressed air (0.2min), places in box; obtains new casting (0.2min),cleans machine with compressed air (0.2min),locates casting in fixture; starts machine and auto feed(0.2min), finishing the casting face by milling machine (0.8min) Suggest some alternative for improving machine and worker utilization.	10	CO6	BT4																								
Q.2	An assembly line for the assembly of an ordinary two pin electrical socket consists of a work bench where the operator places the ceramic body near the workplace, positions two brass hollow pins in slots provided in the ceramic, puts plastic cap cover, screws in set screws at the center with a screwdriver and sets aside the assembly in a pan. The body (base), hollow pins, plastic cap and central screw are available to the operator in tote pans. • Draw a neat sketch of the layout of the workplace you propose for the operator • Make a two-handed process chart for the assembly. • Suggest improvements in the method and in product development for ease in assembly	10	CO5	BT6																								
Q.3	Explain in detail: 1. Cyclegraph 2. Chronocyclegraph 3. Simo chart	10	CO2	BT2																								
Q.4	List down various principles of motion economy regarding use of human body and explain their significance in design of work station layout	10	CO4	BT3																								
Q.5	Explain along with symbols any ten therbligs and their use in micro motion study.	10	CO2	BT3																								
Q.6	A study was conducted on a job consisting of 3 elements. The stop watch reading of four cycles using cumulative timing methods are as: Element Stop watch reading in minute: <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>X</td><td>10</td><td>54</td><td>102</td><td>145</td><td>189</td></tr><tr><td>Y</td><td>25</td><td>70</td><td>116</td><td>159</td><td>194</td></tr><tr><td>Z</td><td>45</td><td>91</td><td>135</td><td>179</td><td>215</td></tr></table> The rating factors were estimated at 90,100 and 100 for the three elements. The relaxation allowances are 10 %.Calculate standard time for the job.		1	2	3	4	5	X	10	54	102	145	189	Y	25	70	116	159	194	Z	45	91	135	179	215	10	CO7	BT4
	1	2	3	4	5																							
X	10	54	102	145	189																							
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Z	45	91	135	179	215																							
Q.7	Write brief notes on any two of the following 1. Types of elements 2. Fly back and cumulative timing methods 3. Factors affecting rate of working	10	CO6	BT2																								

Bloom's Taxonomy (BT)

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BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

At the end of course, student will be able to;

CO 1 Apply basic concepts of productivity and quality of life.

CO 2 Recognize the impact of human factor at workplace for productivity improvement.

CO 3 Calculate productivity of an industry.

CO 4 Apply ergonomics in designing of different products for human comfort at workplace.

CO 5 Learn to implement method study technique in industries.

CO 6 Evaluate the percentage utilization of manpower and machines in industries.

CO 7 Estimate time standards for different processes.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded
Department of Production Engineering
Academic Year: 2025-26 (semester I)
End Term Examination

Class: Final Year B.Tech.	Course Name: Operations and Materials Management	Course Code: PCC-PE401
Date: 27.11.2025	Time: 13:30 to 15:30	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.1	Solve the following questions (Any Four)	Marks	CO	BT
a	Explain the terms: Bottleneck and Constraint.	05	CO4	BT2
b	Why is it importance to manage inventory and operating expenses?	05	CO4	BT1
c	Differentiate between Push & Pull production system.	05	CO3	BT2
d	Explain the Five 'R' s of purchasing.	05	CO2	BT2
e	What is supply chain management? Explain benefits of SCM.	05	CO5	BT1
Q.2	Solve the following questions (Any Four)			
a	How does bottleneck affect the throughput?	05	CO4	BT4
b	Explain the role of purchasing department in material management.	05	CO2	BT2
c	Explain the role of internet in supply chain.	05	CO5	BT2
d	Explain the process of capacity planning.	05	CO3	BT2
e	Write a short note on following terms: • Just in time. • KANBAN. • KAIZEN. • Zero Defects.	05	CO3	BT1
Q.3	Solve the following questions.			
a	If operated around the clock under ideal conditions, the fabrication department of an engine manufacturer can make 100 engines per day. Management believes that a maximum output rate of only 45 engines per day can be sustained economical over a long period of time. Currently, the department is producing an average of 50 engines per day. What is the utilization of the department relative to peak capacity? Effective capacity.	04	CO2	BT5

b	An oil engine manufacturer purchases lubricants at the rate of Rs. 42 per piece from a vendor. The requirements of these lubricants are 1800 per year. What should be the ordering quantity per order if the cost per placement of an order is Rs. 16 and inventory carrying charges per rupee per year is 20 paise.	04	CO3	BT5																																				
c	The following example will give the clear and wide information about ABC analysis: Perform the ABC analysis on the following samples of drugs: <table><tr><th>Item Name</th><th>Annual Consumption (units)</th><th>Price per unit (Rs.)</th><th>Ranking</th></tr><tr><td>Drug-I</td><td>300</td><td>10</td><td>5</td></tr><tr><td>Drug-II</td><td>2100</td><td>15</td><td>1</td></tr><tr><td>Drug-III</td><td>44</td><td>10</td><td>7</td></tr><tr><td>Injection I</td><td>1100</td><td>5</td><td>3</td></tr><tr><td>Injection II</td><td>40</td><td>5</td><td>8</td></tr><tr><td>Injection III</td><td>200</td><td>100</td><td>2</td></tr><tr><td>Drug-V</td><td>700</td><td>7</td><td>4</td></tr><tr><td>Drug-IX</td><td>80</td><td>11</td><td>6</td></tr></table>	Item Name	Annual Consumption (units)	Price per unit (Rs.)	Ranking	Drug-I	300	10	5	Drug-II	2100	15	1	Drug-III	44	10	7	Injection I	1100	5	3	Injection II	40	5	8	Injection III	200	100	2	Drug-V	700	7	4	Drug-IX	80	11	6	12	CO2	BT5
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=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recognize operation strategy concept.
CO 2	Develop the material management planning for final product.
CO 3	Analyze the different push and pull production system tools.
CO 4	Implement the concept of TOC and MOST techniques.
CO 5	Understand SCM used to improve the Supply Chain performance